PHASE 7

Supporting Services and Testing

Image generation, PDF export, layout, utilities and the test suite

COMICCRAFT • AI COMIC STORY CREATOR • TECHNICAL DOCUMENTATION

1. Supporting Services

1.1 Image Generation — app/services/image_generator.py

A provider-abstraction layer built around an ImageGenerator abstract base class. Each provider implements a common interface, selected at runtime by IMAGE_PROVIDER:

Provider	Description
stable_diffusion	Local Hugging Face Diffusers pipeline (default).
huggingface_api	Hosted Hugging Face Inference API — no local GPU required.
gemini_image	Google Gemini image model, using GEMINI_API_KEY.
placeholder	Explicit opt-in offline provider for UI work; never selected silently.

1.2 PDF Export — app/services/exporters.py

Builds a print-ready comic PDF with FPDF2: title page, per-panel layout with captions, narration and dialogue, and a consistent brand palette (purple/cyan accents on a dark-ink text colour).

1.3 Layout Builder — app/services/layout_builder.py

Merges the outline, the generated story text and the per-panel image results into the final ComicPanel objects consumed by both the Jinja2 preview template and the JSON API.

1.4 Utilities — app/utils/

- validators.py — normalises and repairs AI JSON output (recovers malformed JSON, enforces required fields).
- helpers.py — safe filename generation, path-traversal guards (ensure_within), static URL building, logging setup.

2. Testing

The automated test suite uses pytest with FastAPI's TestClient (httpx) and forces IMAGE_PROVIDER=placeholder so tests never call paid external services.

Terminal

```
python -m pytest -q                # full suite
python -m pytest tests/test_services.py -v
```

Coverage includes:

- Homepage rendering, health endpoint and OpenAPI schema
- Pydantic request validation and error envelopes
- Path-traversal protection on the /download route
- AI-output normalisation and JSON recovery
- Layout building, image generation and PDF export